Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets with Solutions #1

[1] Consider an experiment with possible outcomes $\{1,2,3,\cdots\}$ If we assign probabilities

$$P[k]=\frac{1}{2^k},\quad k=1,2,3,\cdots$$

to the outcomes, will these probabilities sum to one? Can you have an infinite number of outcomes, but still assign nonzero probabilities to each outcome?

SOLUTION: Yes, because $\sum_{k=1}^{\infty} P[k]=\sum_{k=1}^{\infty}\left(\frac{1}{2}\right)^k=\left\{\sum_{k=0}^{\infty}\left(\frac{1}{2}\right)^k\right\}-\left(\frac{1}{2}\right)^0=\frac{1}{1-\frac{1}{2}}-1=1.$

Note that we have used $\sum_{k=0}^{\infty} x^k=\frac{1}{1-x}$ if $|x|<1$

For the second part, the answer is yes. However, the numbe of outcomes must be countable and probabilities can't be equal. ∎

[2]. Let $P[k]=\binom{N}{k}p^k(1-p)^{N-k}$. Prove that

$$\sum_{k=0}^{N} P[k]=1.$$

Hint: Firs prove the binomial theorem $(a+b)^N=\sum_{k=0}^{N}\binom{N}{k}a^k b^{N-k}$ by mathematical induction. Use Pascal's "triangle" rule

$$\binom{M}{k}=\binom{M-1}{k}+\binom{M-1}{k-1},\qquad \binom{M}{k}=0,\ k<0 \text{ and } k>M$$

SOLUTION:

Mathematical induction: $N=1\Rightarrow$ $a+b=\sum_{k=0}^{1}\binom{1}{k}a^k b^{1-k}=\binom{1}{0}a^0 b^1+\binom{1}{1}a^1 b^0=b+a$

Now assume that the result is true for N and consider N+1.

$$(a+b)^{N+1}=\sum_{k=0}^{N+1}\binom{N+1}{k}a^k b^{N+1-k}=\left\{\sum_{k=0}^{N}\binom{N+1}{k}a^k b^{N+1-k}\right\}+\binom{N+1}{k}a^{N+1}b^{N+1-(N+1)}$$

$$=\left\{\sum_{k=0}^{N}\left[\binom{N}{k}+\binom{N}{k-1}\right]a^k b^{N+1-k}\right\}+a^{N+1}$$

$$=b\sum_{k=0}^{N}\binom{N}{k}a^k b^{N-k}+\sum_{k=0}^{N}\binom{N}{k-1}a^k b^{N+1-k}+a^{N+1}$$

$$=b\sum_{k=0}^{N}\binom{N}{k}a^k b^{N-k}+\sum_{k=1}^{N}\binom{N}{k-1}a^k b^{N+1-k}+a^{N+1}$$

$$=b\sum_{k=0}^{N}\binom{N}{k}a^k b^{N-k}+a\left[\left\{\sum_{\ell=0}^{N-1}\binom{N}{\ell}a^{\ell}b^{N-\ell}\right\}+a^N\right]$$

$$=b(a+b)^N+a\sum_{k=0}^{N}\binom{N}{k}a^k b^{N-k}$$

$$=b(a+b)^N+a(a+b)^N=(a+b)^N[a+b]=(a+b)^{N+1}\ //$$

Now, $\sum_{k=0}^{N} P[k] = \sum_{k=0}^{N} \binom{N}{k} p^k (1-p)^{N-k} = (p+[1-p])^N = 1^N = 1.$ ∎

[3] If $\int_a^b p_T(t)dt$ is the probability of observing T in the interval $[a,b]$, what is $\int_{-\infty}^{\infty} p_T(t)\,dt$?

SOLUTION: $\int_{-\infty}^{\infty} p_T(t)\,dt$ represents the probability that T is in the interval $(-\infty,\infty)$. Hence, it must be equal to 1 since the interval is the set of all possible outcomes ∎

[4] Let $p_T(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(t-7)^2}$, $-\infty < t < \infty$

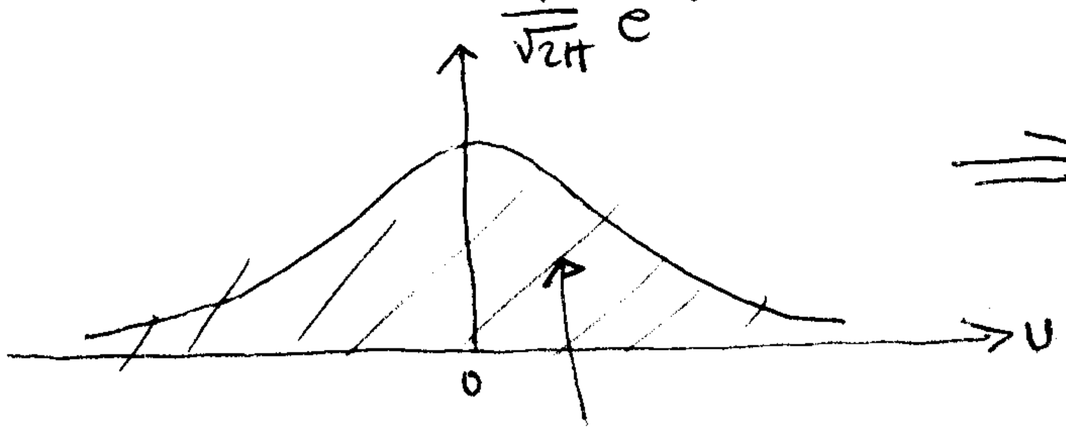
What is the probability of $T > 7$? (Hint: Observe that $p_T(t)$ is symmetric about $t=7$

SOLUTION: $P[T>7] = \int_7^{\infty} p_T(t)\,dt = \int_7^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(t-7)^2} dt$

Let $u = t-7$. Then,

$P[T>7] = \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du$

However, $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du = \underbrace{\int_{-\infty}^{0} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du}_{I_1} + \underbrace{\int_{0}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du}_{I_2}$



area under the curve is 1.

$\Rightarrow$ By symmetry, $I_1 = I_2$

Hence, $1 = \frac{1}{2} + \frac{1}{2}$

$P[T>7] = \int_7^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(t-7)^2} dt$

$= \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du$

$= \frac{1}{2}$ ∎